

ECEN 454 - Lab2: Cadence Custom layout: Design Rules, Extraction, and Verification

1. Introduction

In this lab you will create the physical layout of all the logic gates (Inverter, NAND2, and XOR2) that you used in the previous labs. This lab manual contains instructions to draw the schematic and layout of a basic inverter circuit. You will also perform extraction of the layout parameters and verifying if the layout drawn is equivalent to the schematic (of the corresponding gate) through an LVS (Layout versus schematic) check. You would have to implement the same for the other required gates. Make sure you have attached the gdpk045 technology file to your library before you begin the lab. TInverter circuit

a. Schematic view

- In the Library Manager window, create another library named “Design2”, and do the same steps as in lab 1. Choose “Attach to an existing technology library” and click OK. Then choose “gdpk045”. Then File > New > Cell View. Create a cell named "inverter" in the “Design2” library with its "schematic" view as seen in Figure 1.

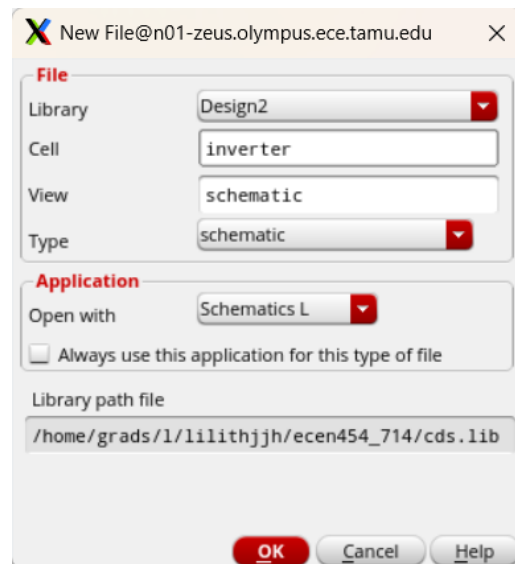


Figure 1. Creating a new schematic cell view of inverter

- After the schematic window opens, we wish to add two transistors to form an inverter circuit. Select Create > Instance (keyboard shortcut: i) to open “Component Browser” window. (If you cannot use the shortcut, or “Component Browser” window won’t come up, just ask TA. This implies your cadence setup has some problem.) Make sure that “gdpk045” is selected in Library list as shown in Figure 2. Let us add a PMOS transistor first. Please select “mos” in the list, and you will find “pmos1v” from the list of “mos”. Switch to the schematic window again and click a position where you would like to add a PMOS transistor. In a similar way, you can add NMOS transistor in your schematic from Component Browser by selecting “nmos1v” from the list of mos. Press ESC to escape from Add Instance mode.
- Connect components with wires by selecting Add > Wire or use the “w” shortcut.

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Figure 2. Adding a PMOS transistor from Component browser

- Add pins and give proper directions and names. Select Create > Pin or use the “p” shortcut. Please be very careful when you put name and direction of each pin since those must be consistent between schematic, layout, and symbol. For the power pins, DO NOT use the VDD and GND symbols. Instead, create two “inputOutput (Direction of pin)” pins with the names VDD or GND.
- To change the properties of a component, click Edit > Properties > Objects or use the “q” shortcut. Please try to change the width of transistors. To give similar logical strength to both rising and falling output, the width of a PMOS transistor will be at least twice larger than that of an NMOS transistor.
- When completed, please check your schematic with a sample circuit given in Figure 3.

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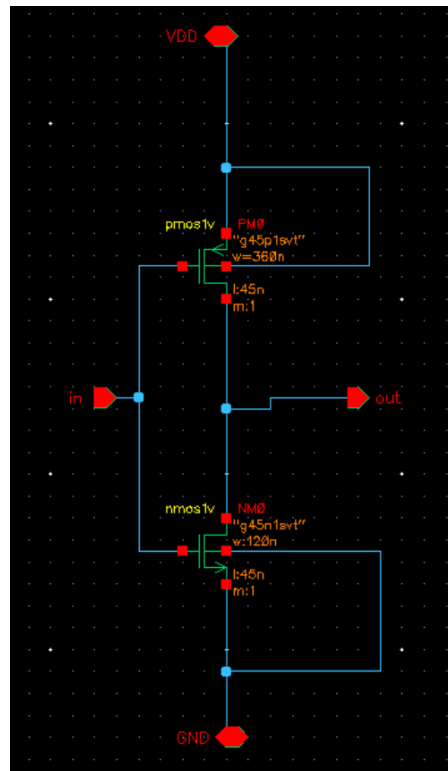


Figure 3. Inverter schematic

- Select File> Check and Save to verify your design and save it. Make sure you don't see any errors or warnings.

b. Symbol view

In the schematic view window, please select Create > Cellview > From Cellview. A window as Figure 4 will open. Press OK.

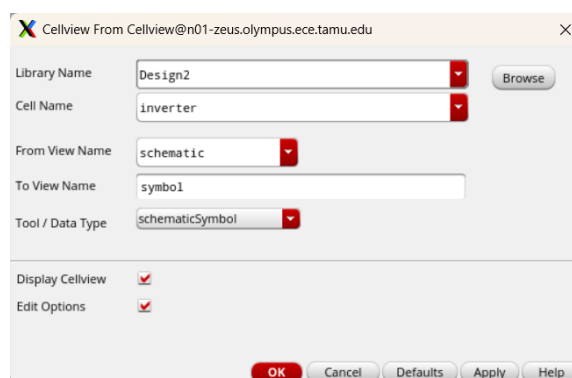


Figure 4. Create symbol from schematic view

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- The symbol which is initially created will look like Figure 5. Please modify the symbol so that it resembles a general symbol of inverter as given in Figure 6 (See appendix). Then save it. **It's also ok if you change the port name and leave the shape as the default one. You don't need to draw hours on it.**

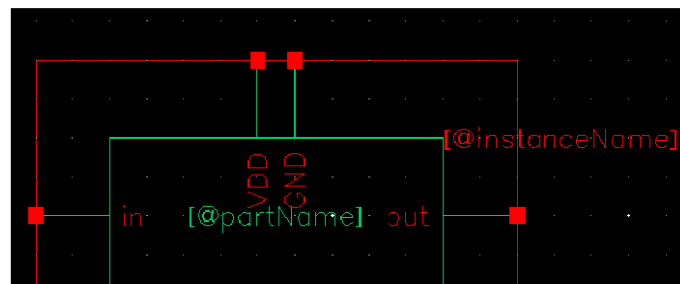


Figure 5. Default symbol of inverter

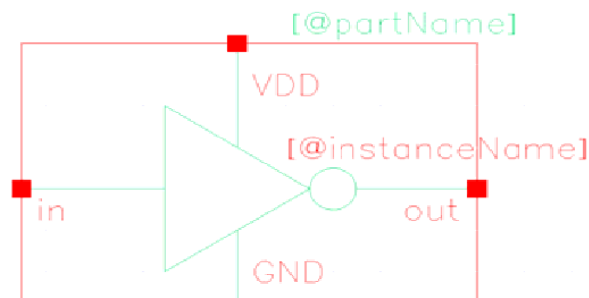


Figure 6. Complete inverter symbol

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c. Layout view

- After the schematic and symbol of an inverter circuit are done, next step is to design layout. Create a new "layout" view by File > New > Cell View, and select "layout" as the type, as shown in Figure 7. The Virtuoso layout editor window and the layer selection window (LSW) window will pop up.

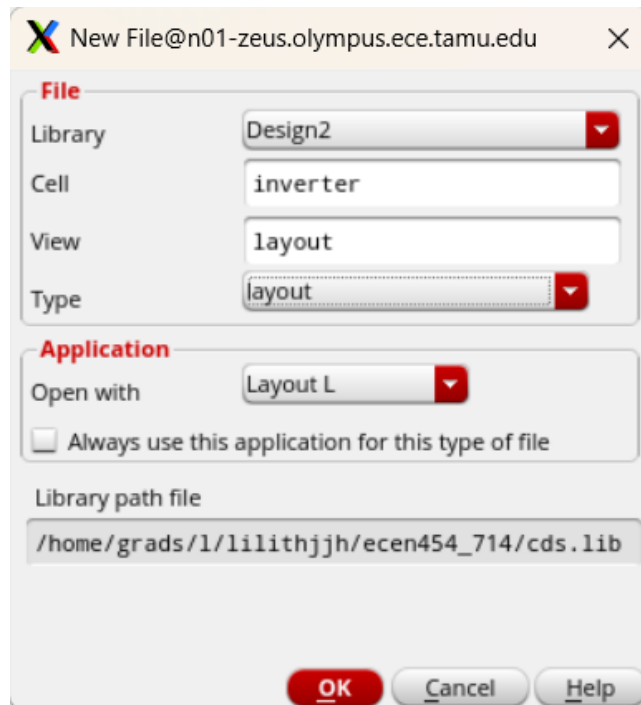


Figure 7. Creating a new layout view for the inverter

- You need a proper display setting to see the names of your pins. Select Options > Display and in "Display Controls" darken "Pin Names".

To make grid snapping better you may want to change the snap spacing to 0.01. Go to Options -> Display or use hotkey to view the display options. Under X Snap Spacing & Y Snap Spacing, set these both to 0.005 for more precise layout positioning. **NOTE: Incorrect Grid Spacing can cause DRC errors which may have you redo your entire design.**

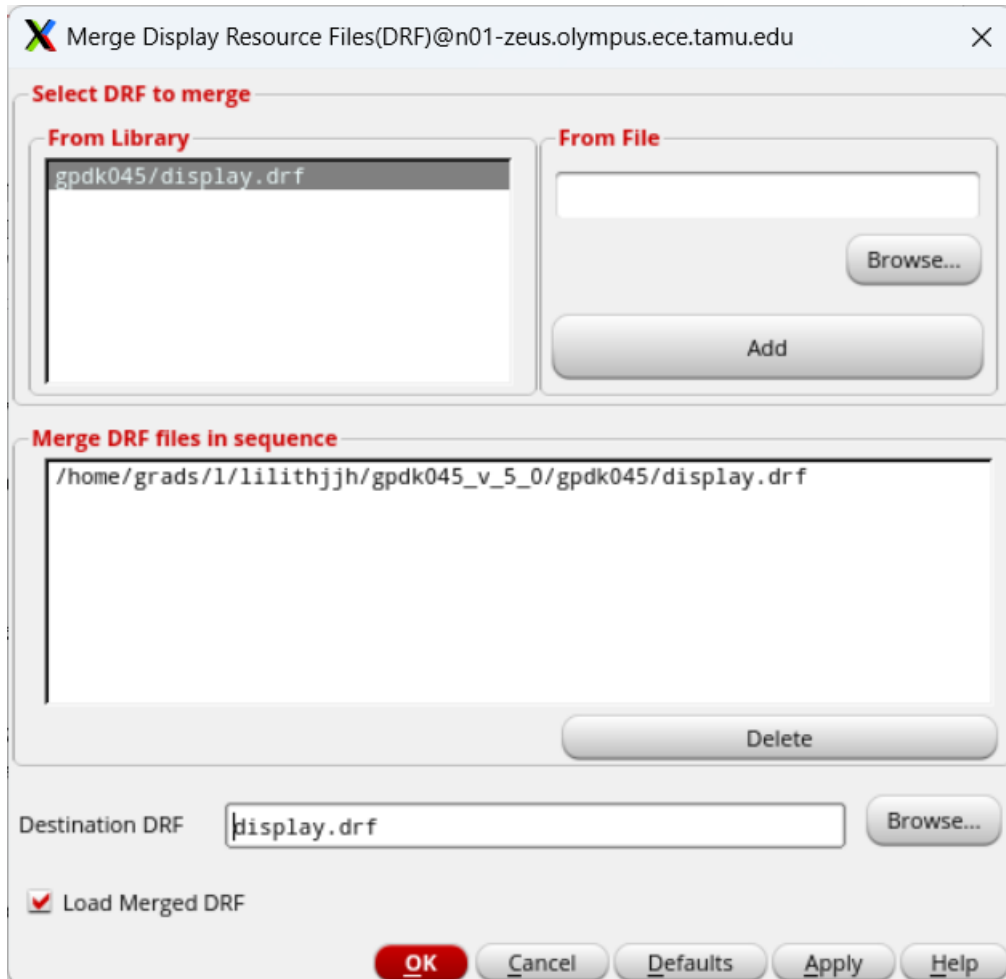
When you draw the layout, it may pop out a window says "There are undefined packers found."

You can open the CIW, Go to **Tools > Display Resource Manager**.

Click "Merge", make sure you select the file under the folder "gpd045", and name your new destination file, typically display.drf

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Then click “OK”, it should be all set.



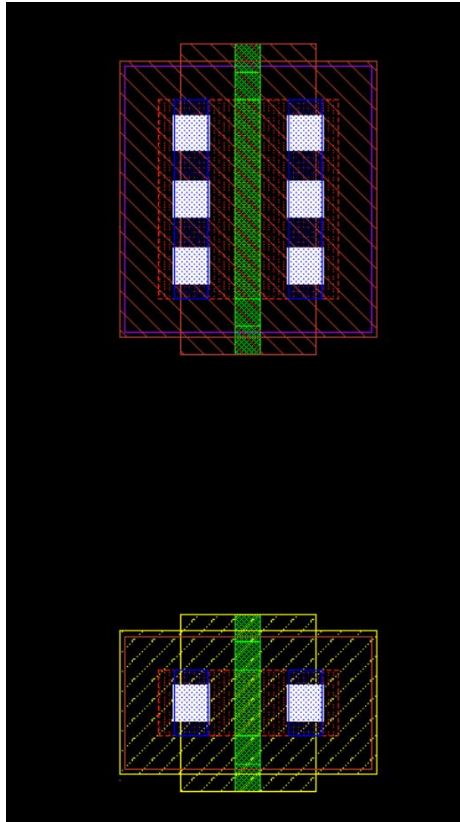
On the left-hand side, there is a pane that contains all the layers that would be used to draw the layout. To create the inverter layout, let us first instantiate the PMOS transistor. To do this, press hockey i or click on Create -> Instance. In the create instance window, enter the following:

- Library: **gpdk045**
- Cell: **pmos1v**
- View: **layout**
- Length: **45n**
- Finger Width: **360n**
- Total Width: **360n**
- Fingers: 1
- S/D Metal Width: 60n

Place the instance in the layout window. Similarly, instantiate a *nmos* transistor from the same library and

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place below the pmos transistor in the layout window. The layout should look similar to one shown below.



Press ShiftF/ControlF to toggle views.

The green rectangle (poly layer) in the middle of each transistor is the gate of the transistor. The white filled rectangles on either side of the gate are vias and can be used interchangeably as source/drain terminals of the transistor. Now, we connect the gates of the pmos and nmos transistors using poly layer.

- Left click on the poly layer in the layers pane.
- Then switch back to the layout window and press rectangle hotkey *r*. The rectangle window appears. Check the box to enable smart snapping.
- Switch back to the layout window and left click at the left bottom corner of the rectangle you want to draw.
- Then left click at the top right corner of the rectangle you want to draw.

Most hotkeys used in the schematic editor work in the layout editor too. After drawing a rectangle, you might want to take a few minutes to familiarize yourself with manipulating the rectangle shape and size.

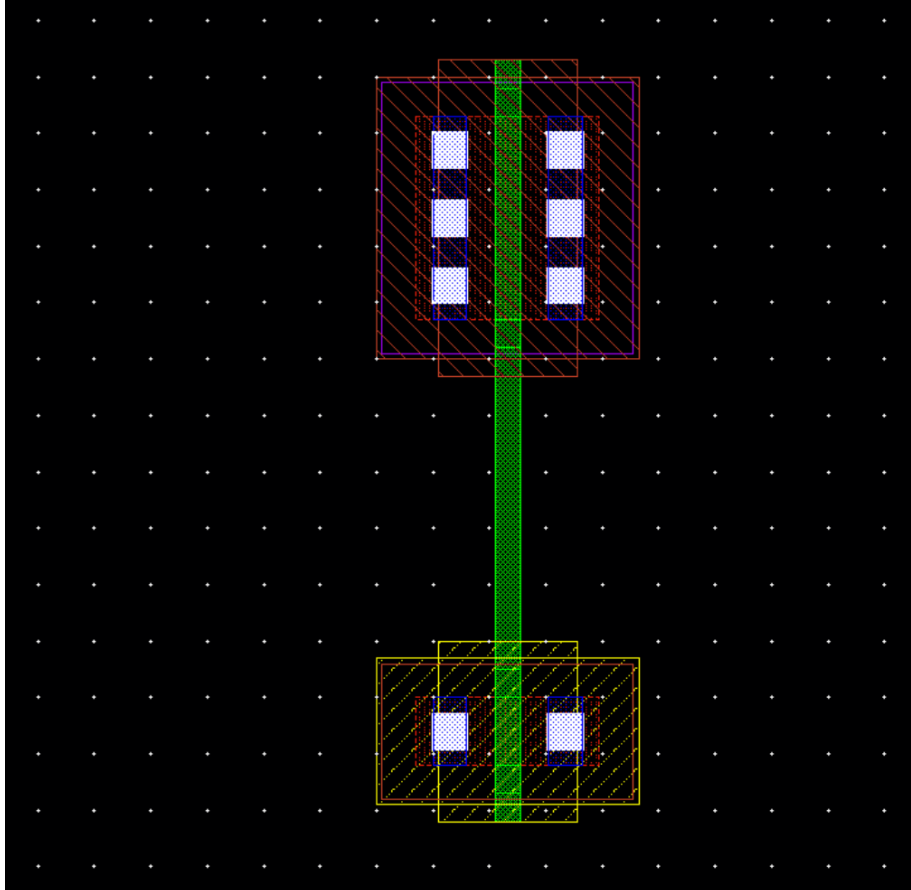
Try using different hotkeys such as

- m: move
- s: stretch
- p: path
- o: via

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- ctrl/shift z: zoom in/out
- f: Fit to Window
- k: ruler
- u: undo
- shift u: redo

To cancel the current command, press ESC. At the end of this, the gates should be connected, as shown below:

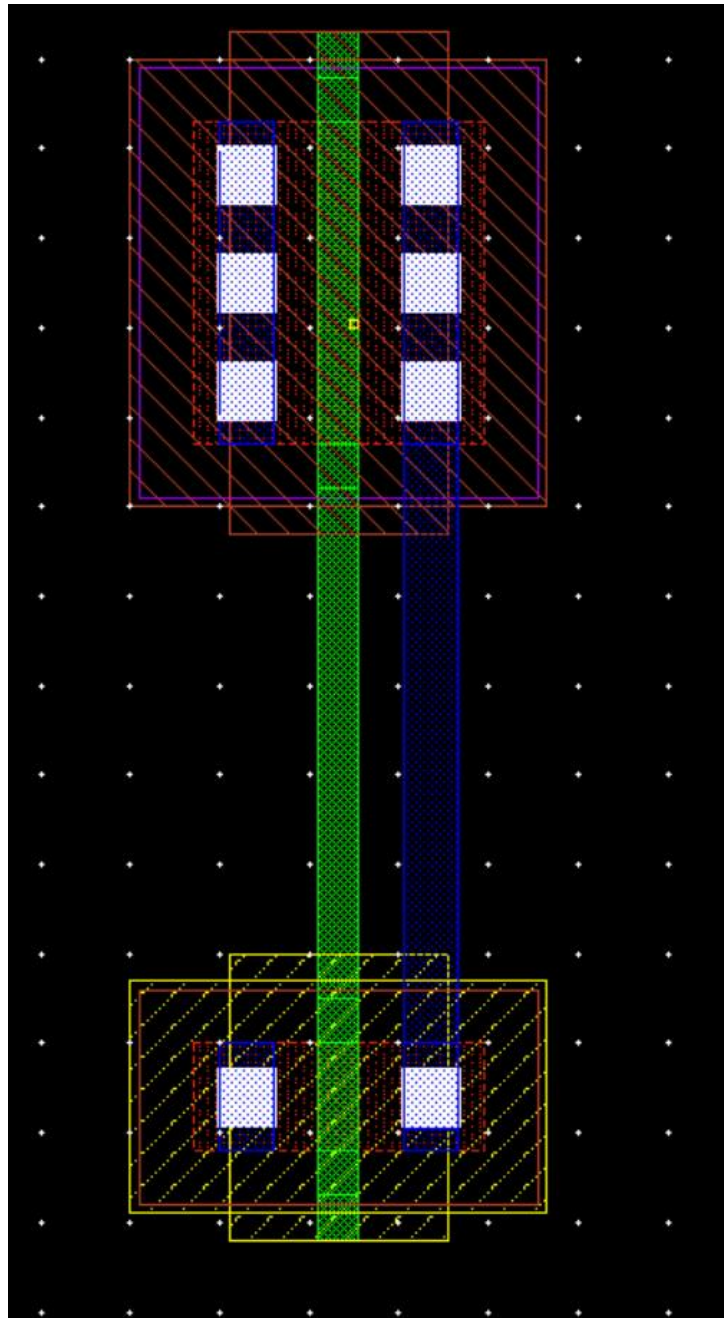


As we can see, the width of the poly rectangle may not perfectly aligned with the gate of the two transistors.

To fix that select the drawn poly & press the hotkey Q. Change the rectangles **Width** to 0.045 & press Apply to see the changes. Once, you're happy with the changes, press OK to close the properties window.

Then create a metal 1 wire (using rectangle or path) to connect the drains of the two transistors (connect the two vias, as shown below). Use more rectangles to draw out the wires.

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The poly layer currently is connected to our gates. We want the *in* pin to connect to the poly layer. Instead of directly connecting to the poly layer, lets use a metal as it will be more conductive than poly.

To transfer between two layers or more layers, use a via. GO to Create -> Via
Set the following parameters for a single metal 1 to poly via.

- Mode: Single
- Via Definition: M1_PO (Metal 1 to Poly)
- Rows: 1
- Columns: 3

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The screenshot shows the 'Via Definition' dialog box in Cadence. The 'Mode' section has 'Single' selected. 'Compute From' has 'Row/Col' selected. 'Constraint Group' is set to 'Same as Wire'. The 'Via Definition' section shows 'M1_PO' as the definition and 'Standard Via / gpdk045' as the technology. 'Rows' is 1 and 'Columns' is 3. The 'System' tab is active, showing 'Justification' as 'centerCenter' with X and Y offsets of 0. The 'Cut' section shows 'Size' and 'Spacing' for both X and Y as 0.06, with 'Cut Class' set to 'None'. The 'Enclosures' section has 'Compute' checked. At the bottom, there are 'More Options' buttons: 'Rotate', 'Sideways', and 'Upside Down', along with 'Hide', 'Cancel', and 'Help' buttons.

Mode: ☒ Single ☐ Stack ☐ Auto ☐ Fast Edit

Compute From: ☒ Row/Col ☐ Width/Height ☐ Shape(s) ☐ Drawn Area

Constraint Group: Same as Wire ☐ Use All Vias

Via Definition: M1_PO Standard Via / gpdk045

Rows: 1 Columns: 3

System: ☒ User defined ☐ Cut pattern ☐ Array pattern

Justification: centerCenter 0 0

Cut:

Size X: 0.06 Y: 0.06 Cut Class: None

Spacing X: 0.06 Y: 0.06

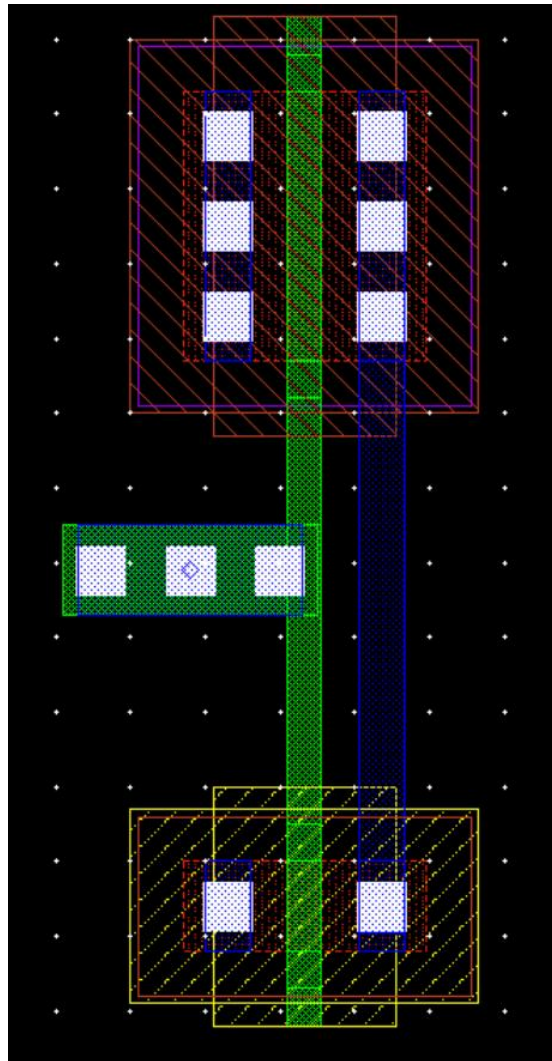
Enclosures: ☒ Compute

More Options:

Place the via so that the right most via is on top of the drawn poly layer to make a connection as shown below.

Save the layout with File -> Save.

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Running DRC

DRC stands for Design Rule Check & it make sures that the layout is following the rules of the constraints from the foundry. In other words, a passing DRC means that there are 0 errors & that the design is up to code for the foundry.

It is best practice to run DRC checks often rather than at the end of finishing the layout so that you can catch errors early instead of later where you possibly may have to change your entire layout.

To run a DRC, go to Pegasus -> Run DRC And then do the following setup:

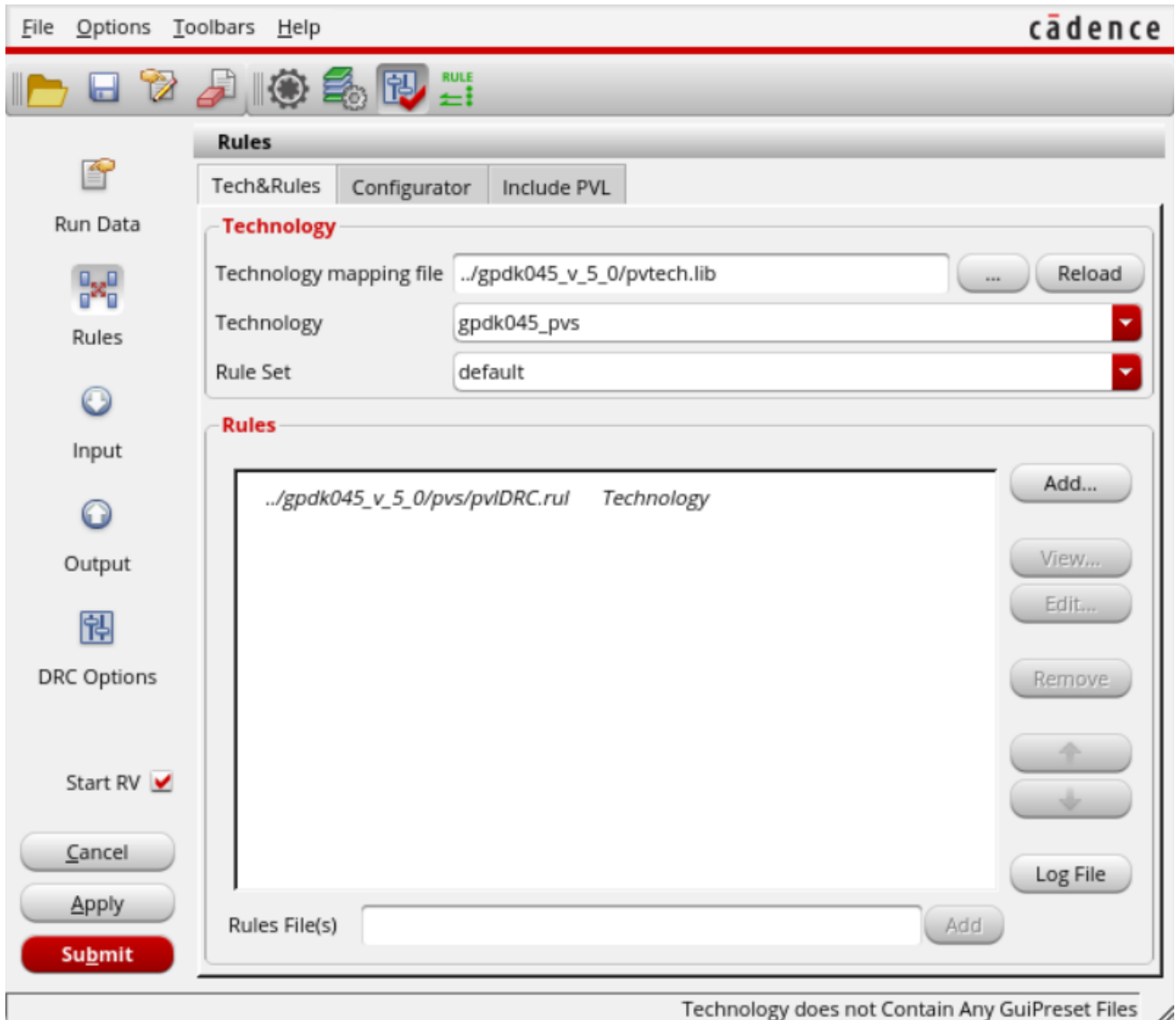
Run Data

- Run Directory: Create a folder names “DRC” and select it.

For the rules, click the icon “...” on the right side of the technology mapping file, select the file pvtech.lib under your unzipped gpd045 file.

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Select gpdk045_pvs as your Technology. Leave the rest as default. Click the red button “Submit”.



The Pegasus Results Viewer should automatically open when DRC has completed. In the Pegasus Results Viewer, if there's no error, it should be empty. Otherwise, you can see a number of errors that pertain to the layout currently. Double clicking on a rule error will zoom into the error on the layout. This is very useful for debugging where errors are & fixing them.

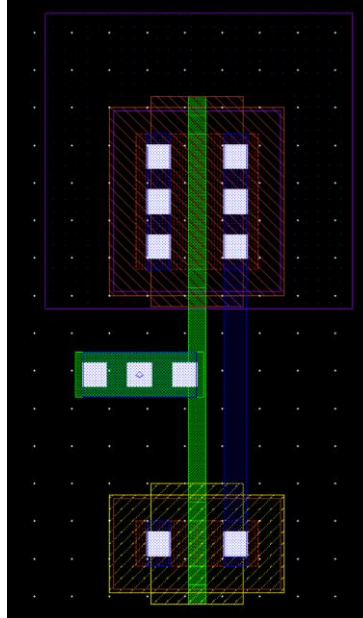
Close the DRC Result Viewer & the report viewer by clicking on the Close Report button in the Pegasus Reports window. We will fix these errors as we continue. You can reopen the report very easily by going to Pegasus -> Recent Runs -> drc.

There are nwell errors that specify that the minimum nwell area is $\geq 0.18\text{m}$ & that the minimum nwell

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width is $\geq 0.3\mu\text{m}$.

Back in the layout we can see that Nwell is used for the pmos transistor. Take a rectangle & draw over the nwell so that it includes the entire pmos transistor within it.

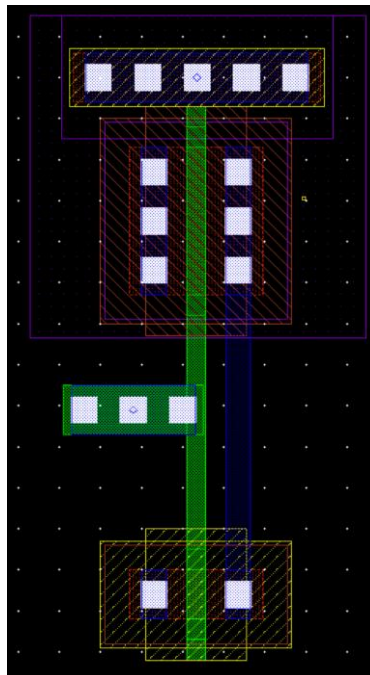


Create another via from M1 to NWELL that have 5 columns and 1 row. Place the via above the pmos transistor.

This via is going to be used for connecting VDD to the nwell. We use 5 columns of vias to ensure better contact although 1 column would work.

Connect the metal 1 source of pmos to the via to create a path for vdd to connect through.

By now, your layout should look similar to this:

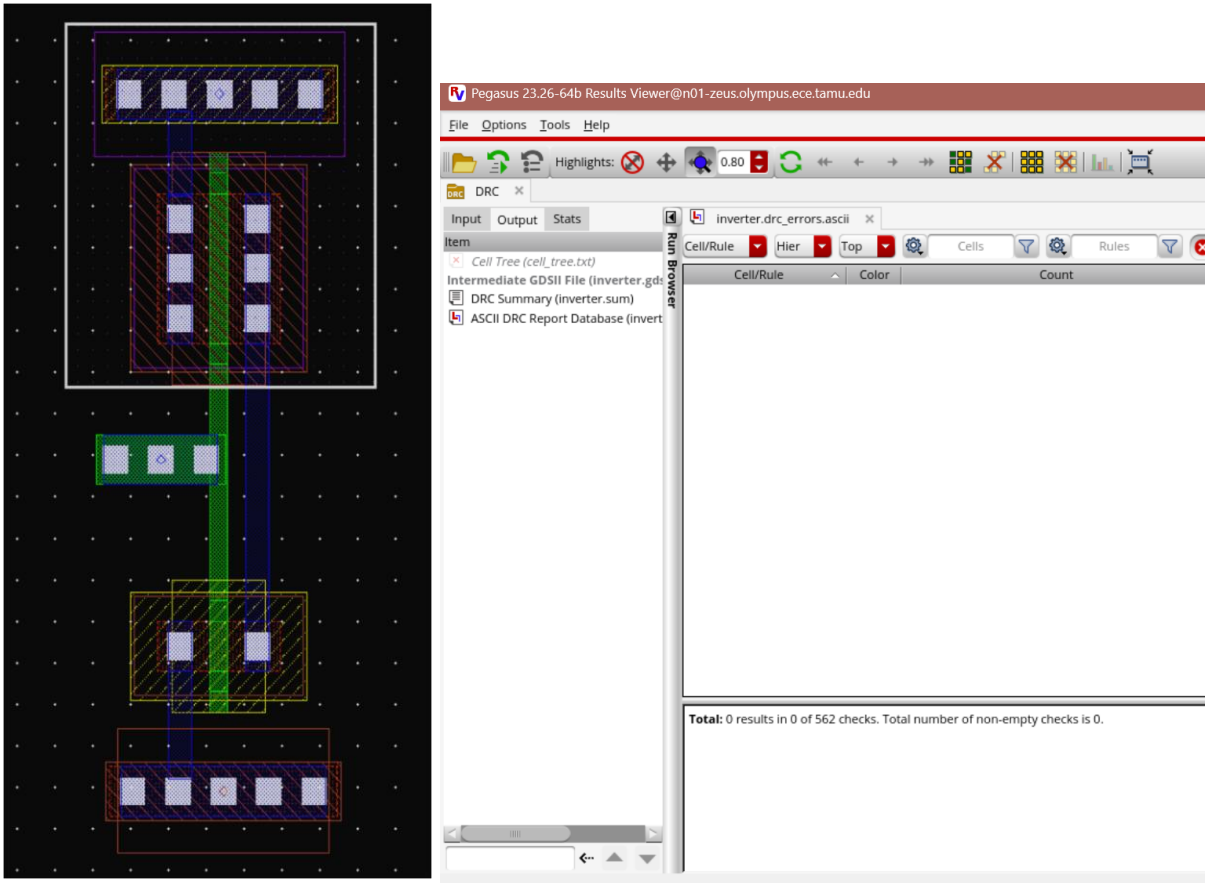


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Similarly, add a via from metal 1 to p-substrate (M1_PSUB) with 5 columns, underneath the nmos transistor.

Connect the source metal 1 of the nmos to the via to create a contact for ground (gnd!) to flow through.

After this, you should pass all the DRC check. Your layout and drc report would look like:



Running LVS

LVS stands for Layout vs. Schematic. It is a check that compares a layout with a schematic & sees if all the nets, inputs & outputs match. An LVS passing means that your layout in ideal circumstances, would perform exactly as your schematic did in simulation.

To run LVS, go to Pegasus -> Run LVS

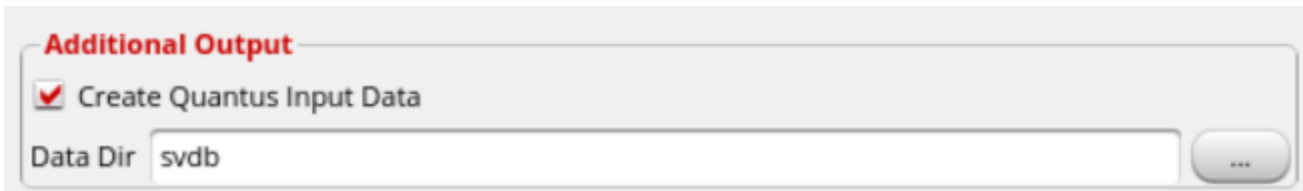
The setup of “Run Data” and “Rules” are the same as DRC’s.

For the input part, make sure you change the “Convert Pin to” to “Geometry+Text”

Convert Pin to ☐ Geometry ☐ Text ☒ Geometry+Text ☐ Ignore

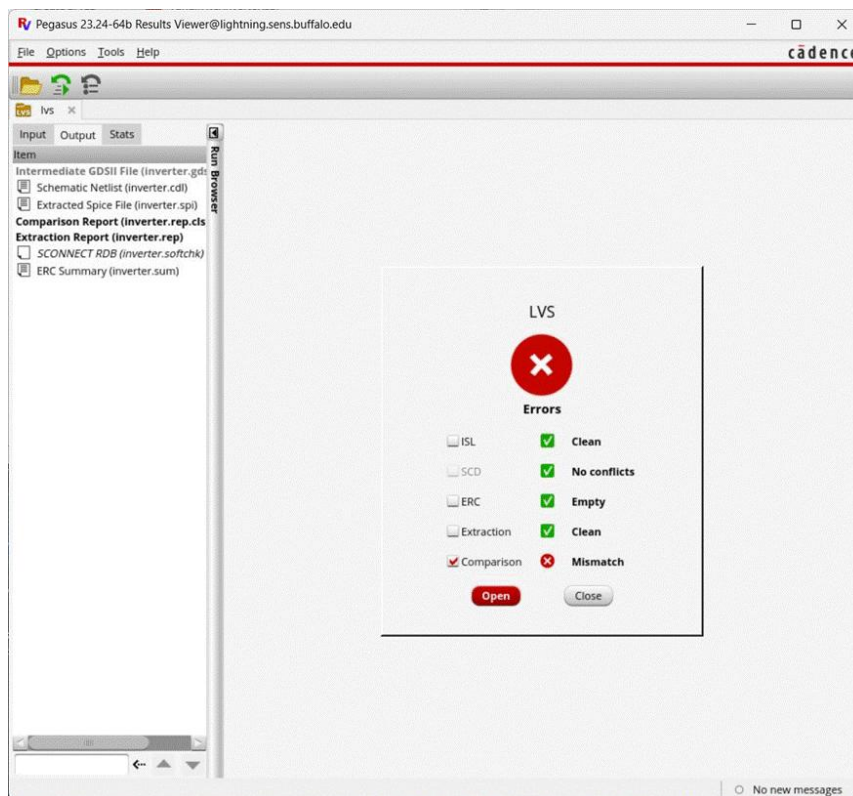
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And in the output part, click on “Create Quantus Input Data”



The click “submit”.

You should see the following window when the LVS job completes:



LVS did not match the netlists so click on Open to view the report.

From the report, click on inverter on the right pane to view the cell specific errors.

We can see from the report that each of the four pins from the schematic were unable to be matched on the layout. Next, we will fix the errors so that LVS passes. Close the report with the Close Report button similar to DRC.

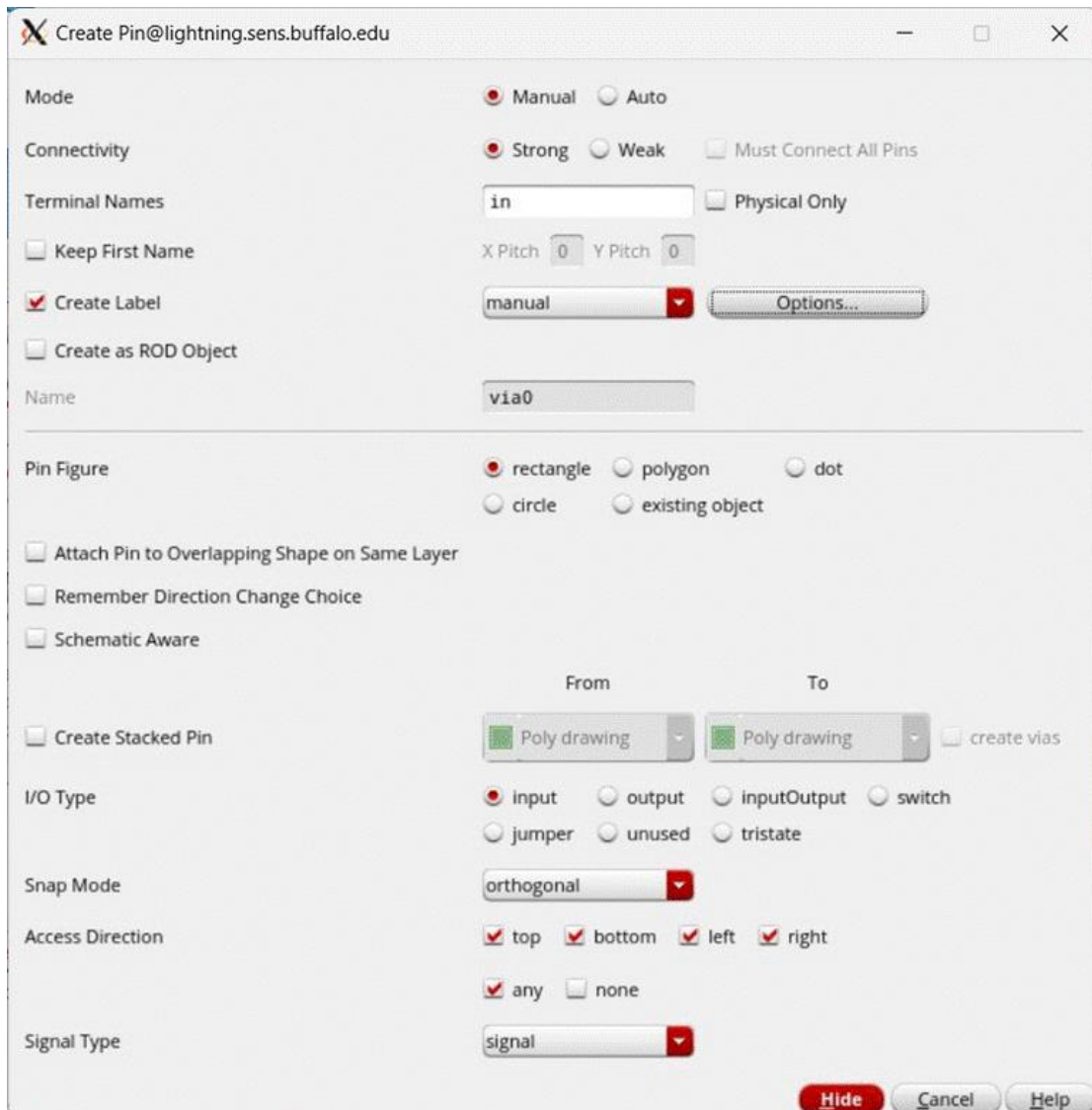
To create a pin on the layout go to Create -> Pin.

In the following window, set the following:

- **in** as the terminal name
- Check the **Create Label** option
- Click on options & check auto under Height. Click OK.
- You can also set the height as you like (0.1) is pretty good for these small devices.
- Select **input** for I/O Type

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Select the metal 1 layer to create the pin on metal 1. Metal is always a good choice for pins as it is the most conductive.



Place the pin on top of the M1_POLY via that connects the two gates of the transistors. Placing the pin is similar to placing a rectangle. Clicking once will start the pin at that corner. Clicking again will create the whole pin. Next, click where you want the pin label to appear. Ideally, pick a spot that only has one layer underneath the pin so that later on there is less confusion as to what the label connects to.

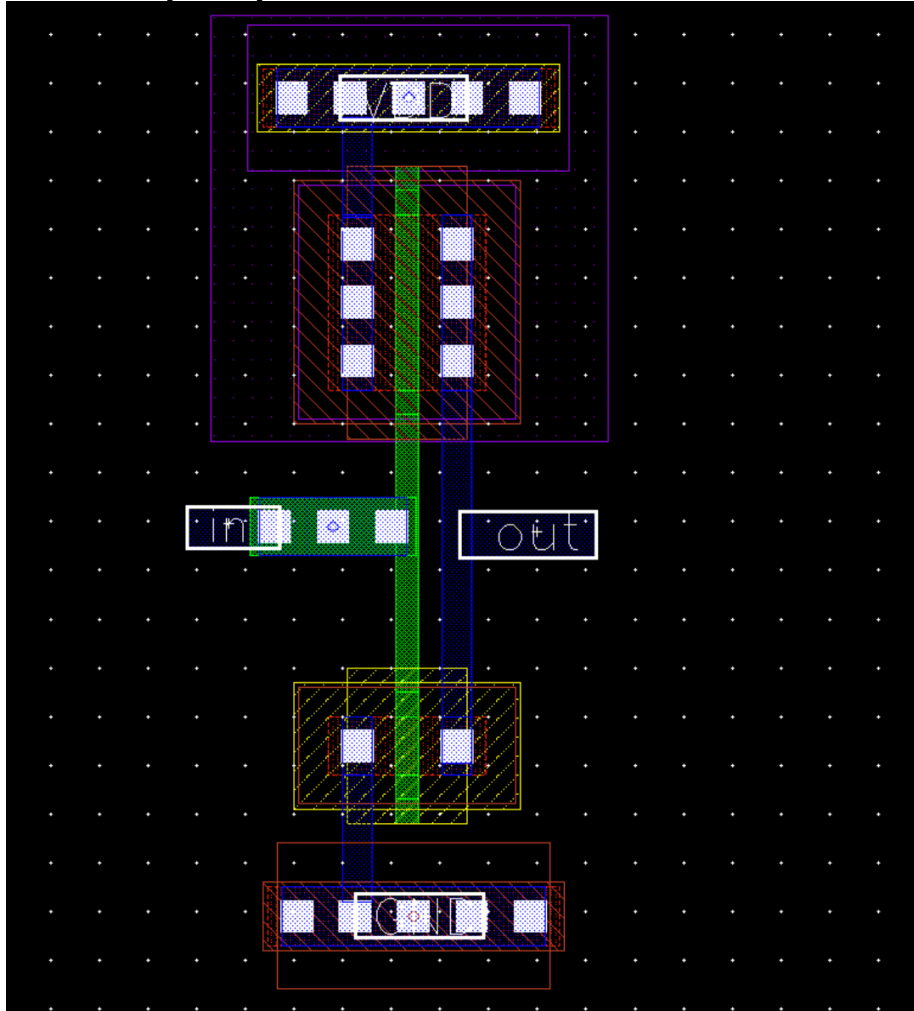
Similarly, create a pin for the **out** pin. This time, set the I/O Type as output. Place this pin on the metal 1 between the two drains of the transistors.

Lastly, create the vdd! & gnd! pins. For pins that have the same properties, you can specify multiple terminal names (separated by spaces) & the tool will allow you to place more pins until the terminal names list is done. Set vdd! & gnd! pins to inputOutput I/O Types.

If the labels of pins are too big you can edit them by clicking the label & selecting Edit -> Basic ->

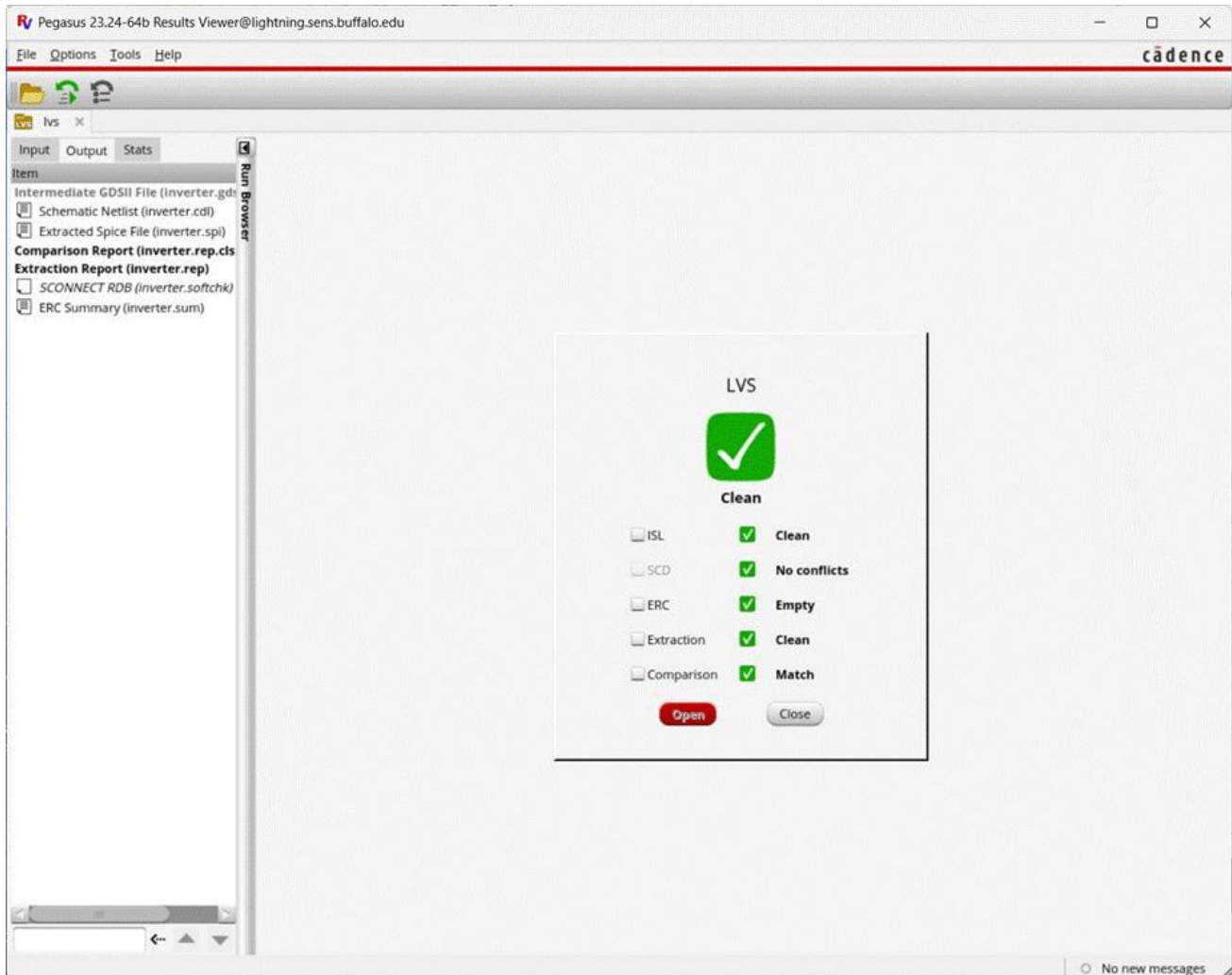
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Properties or using the hotkey q, and change the height of the label from there.
Once the pins are created, your layout should look similar to this:



Run LVS again under Pegasus -> Run LVS & make sure that the form options are correct.
LVS netlist should now match with the following window.

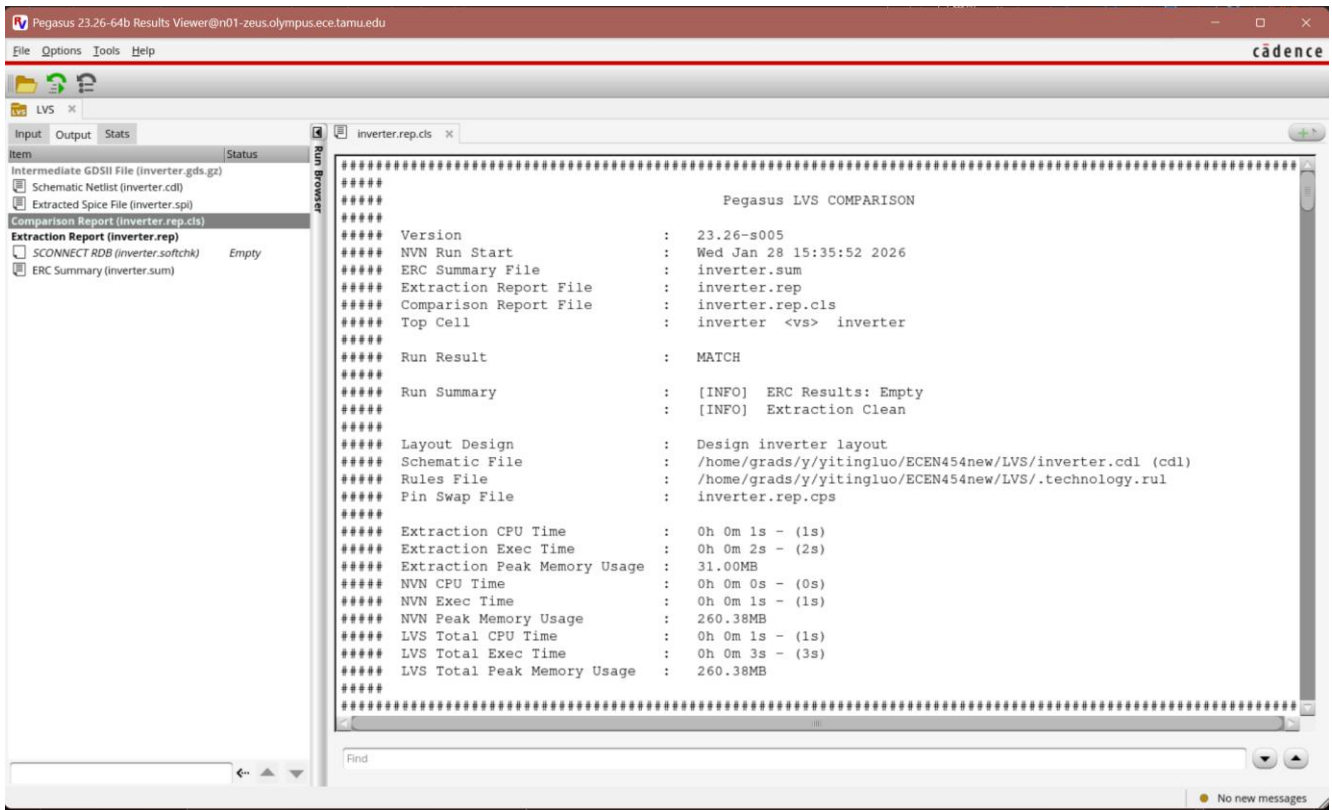
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Make sure you include the Comparison Report for every design cell view you draw when you submit your lab report, this is the only thing to prove you draw a good layout and pass the LVS test.

A clean LVS Comparison Report would look like:

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If the LVS job fails, please check the followings:

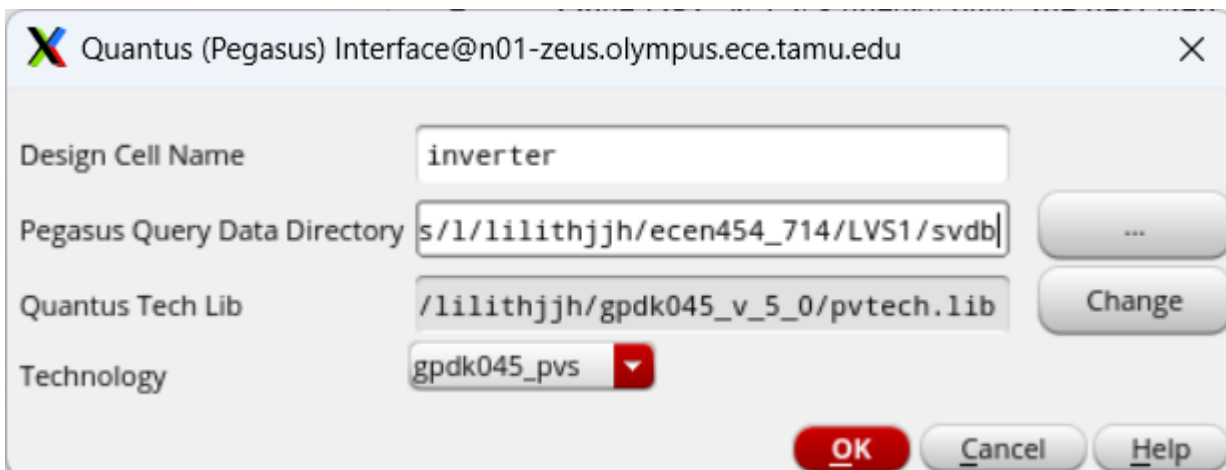
- Have you done with DRC check and Layout extraction before LVS, especially if you made any updates in your schematic or layout view?
- Is your schematic OK? Do you see any errors when you select “check and save” again?

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Parasitic Extraction with Quantus

Once DRC & LVS checks pass, the next step is to do a parasitic extraction of the electrical circuit netlist from the layout.

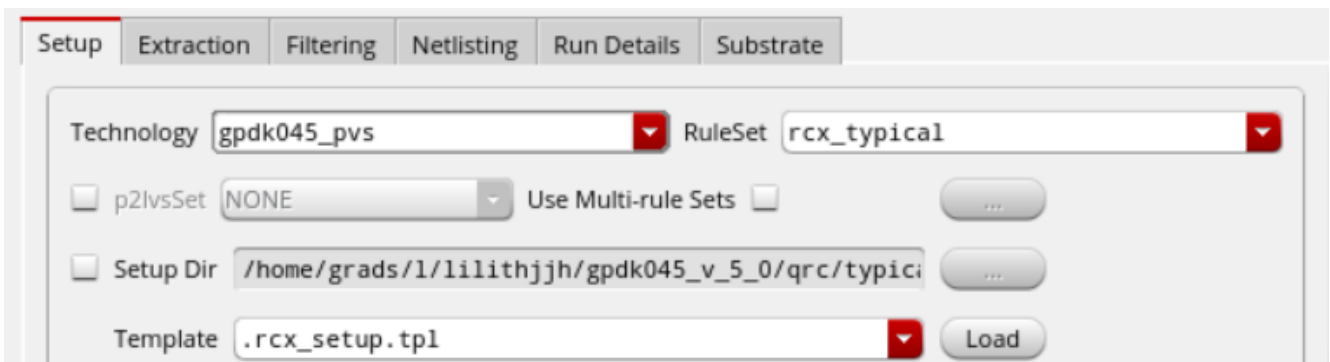
To set up the parasitic extraction, click on Quantus -> Run Pegasus Quantus. The interface window should pop up and be already prefilled from the latest LVS ran as shown below:



Notice that the svdb file as the directory from the LVS Output options earlier. Quantus can only be ran after an LVS of the layout has been ran so keep this in mind.

Type in the cell name you want, for example, inverter. Find the “svdb” file you just created and saved in the previous steps, and click “OK”.

In the setup page, change “Technology” to “gpdK045_pvs”, “Rule Set” to “rcx_typical”. Then roll down to the output part, change it to “Extracted View”, and leave the set as default.



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Output: **Extracted View** (dropdown)

Lib: **design** (text) Cell: **inverter** (text) View: **av_extracted** (text)

Enable CellView Check: ☒

Replace Square Busbit Delimeter: ☐

Parasitic Res Component: **presistor** (text) Prop Id: **r** (text)

Parasitic Cap Component: **pcapacitor** (text) Prop Id: **c** (text)

Parasitic Ind Component: **pinductor** (text) Prop Id: **l** (text)

Parasitic M Component: **pmind** (text) Prop Id: **k** (text)

Then change to the “Extraction” tag, change the extraction type to “RC”, Cap Coupling Mode to “Decoupled”, and set Ref Node to “gnd!”

Setup | **Extraction** | Filtering | Netlisting | Run Details | Substrate

Extraction Type: **RC** (dropdown) Name Space: **Layout Names** (dropdown)

Max fracture length: **infinite** (text) **microns** (dropdown) Temperature: **25.0** (text) **C** (text) **Edit** (button)

Max fracture length by layer: **Edit** (button)

☐ From File: **...** (button)

Cap Coupling Mode: **Decoupled** (dropdown)

Advanced Coupling: ☐ Mult Factor: **1.0** (text)

Ref Node: **gnd!** (text) Ground Substrate Floating Nets: ☐

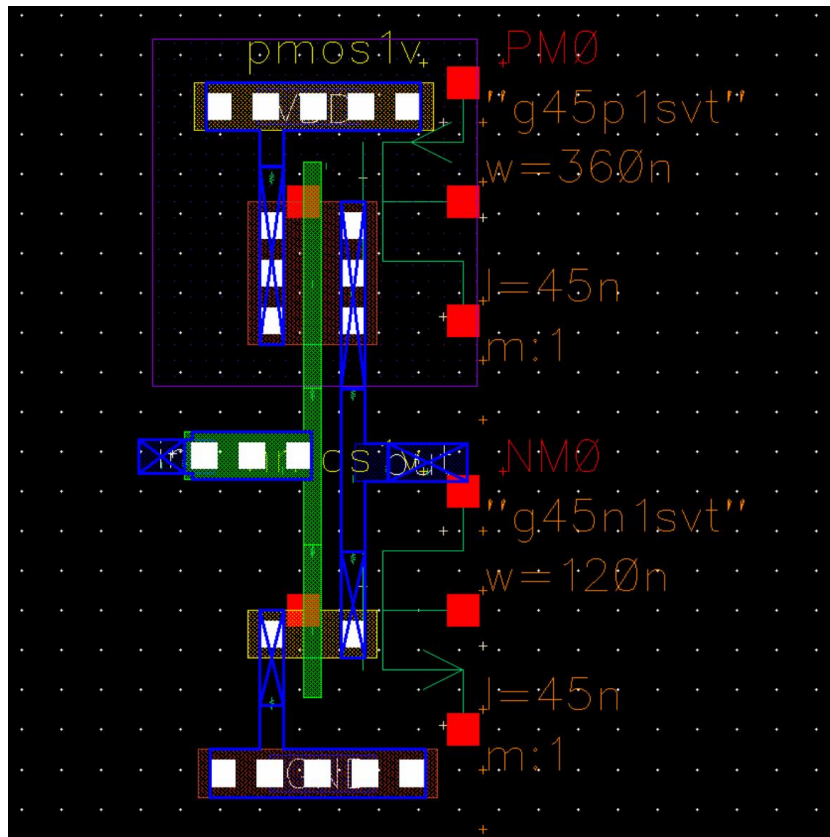
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After the forms parameters are set, click OK to start the extraction. This will take a few seconds.

Once done you should see a popup saying that the extraction was completed & its location.

If there are errors, view the error report or check the CIW for more information.

Once the Quantus run is complete, go to the Library Manager window and open the extracted cell view for the inverter. It should look similar to the window shown below. If it **doesn't** try toggling the view with Shift F/Ctrl F.



The extracted view is for computer to read and use, so you don't need to edit it or try to move anything to make it clear to see when you submit the screenshot.

5. Report Requirements

1. Please apply all the design steps above to Inverter, NAND2, and XOR2 gates.
2. Please submit your lab report **only in PDF** and include screenshots of schematic, symbol, layout and extracted for the "INVERTER", "NAND "and "XOR" gates as well as a screenshot from the LVS comparison report.
3. NOTE:
 - a) When drawing layouts, please refer to the lecture 3 note, p. 17, as a proper example view of your layout design.

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- b) Figure 17 is the schematic of XOR2 gate. This gate has 12 transistors. Figure 17 requires two inverters to generate $\neg A$ and $\neg B$ from input A and B.

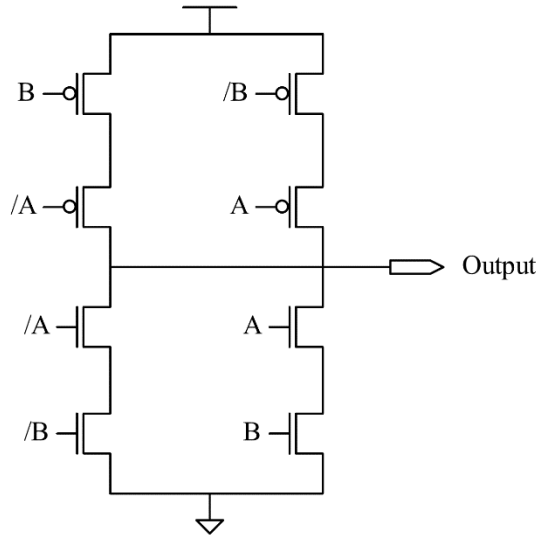


Figure 17. XOR2 gate

- c) Feel free to use other metal layers (e.g. Metal-2) in your layouts, especially for XOR2. In case you need to connect M2 to poly, you need to have two contacts overlapped: **M2-M1 with M1-poly**.
- d) You eventually would need the entire standard gate layouts used (inverter, NAND, XOR) to be of the same height. This will help us make connections easily in future labs when you would use the individual layouts to build the overall circuit. This will have to be taken care of in future lab sessions.

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Appendix: Rubric & Draw custom cellviews

1. Schematic – 0.5 each (inverter, nand, and xor)
2. Symbol – 0.5 each (inverter, nand, and xor)
3. Layout – 1 each (inverter, nand, and xor)
4. Extracted - 0.5 each (inverter, nand, and xor)
5. LVS report – inverter (0.5), nand (1), and xor (1)

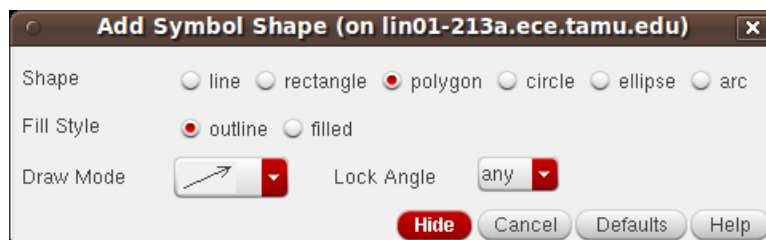
This section is providing instructions to draw custom symbols for gates created as schematic. You may first extract a symbol from your schematic and use this section to update and edit the symbol because the symbol drawn automatically from your schematic will have only rectangle and pins deployed randomly.

As you saw in Lab1 you can create a symbol of any circuit that you design by using the inbuilt option in the Design Window to create a symbol for the current cellview. But you can also create custom symbols for the logic gates rather than the rectangular ones that are created by default. For example you would need to create a triangular symbol followed by a bubble for an inverter.

To create a custom symbol for any design you can create a new cellview “symbol” in the same cell. The following instructions explain how you can create a symbol for an inverter.

In the schematic window, choose Create > Cellview > From Cellview. Type in the cell name of the inverter you have designed as shown below. Click “OK” and a new window will pop-up.

Since you want to draw a triangle, now choose Create> Shape > Polygon and the following figure will pop-up.



Choose Hide and begin at one of the desired vertices of the triangle that you want to draw. Choose all the vertices that you want in the polygon and then click back on the first vertex to complete the figure. Choose Hide and begin at one of the desired vertices of the triangle that you want to draw. Choose all the vertices that you want in the polygon and then click back on the first vertex to complete the figure.

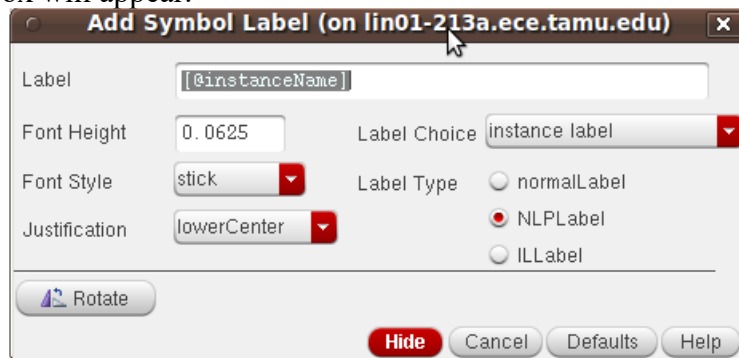
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You can then draw a bubble in front of the triangle by choosing to draw a small circle in from of it. You can click on a point just in front of the triangle and then extend the mouse to indicate the radius of the circle that you want to create.

You should continuously perform “Check and Save” on the symbol by pressing “F8” or choosing “Design > Check > Save” . The symbol will automatically be compared to the schematic in the same cell and at this point will give out some warnings. This is because the tool sees that you have not yet included the names of the pins defined in the schematic in the symbol.

You can add a pin by choosing “Add > Pin” and choose the appropriate direction (input, output or input/output) for it. Add pins for input A, output A', power lines (input/output) VDD and GND. You will see that the pin in this case is slightly different from the one that appears in a schematic. It is a small box with a line attached to it and the pin name on it towards the end. The box is the actual pin and so it must be pointing in the away from the rest of the symbol. Note after you add in all the appropriate pins with proper directions the warnings go away.

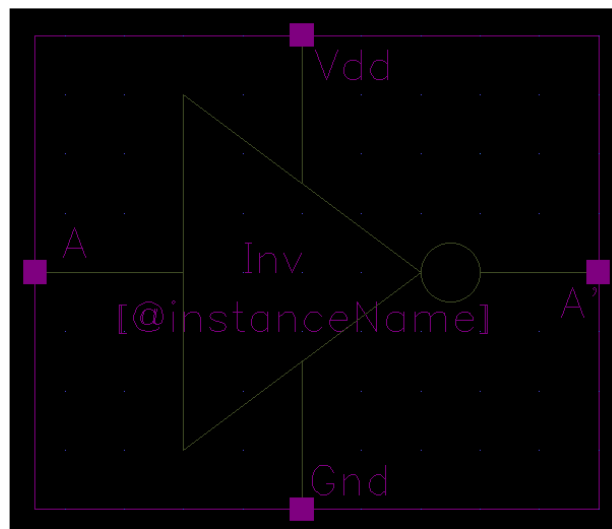
The next step is to create a label for the symbol. You do this by choosing “Add > Label” after which the following dialog box will appear.



The first label you add to the symbol is the [`@InstanceName`] which will be replaced by instance names such I0, I1 etc when the symbol will be used in other designs. This is to differentiate between different instances of the same cell.

The second label you add to the symbol will be the cell name. You do this by choosing “Add > Label”. Then choose Label Type as “NormalLabel” and type in the cell name like “inv” in the Label field.

The final step is to create a selection box for the symbol which defines the area in which the symbol will be selected if you were to click in it. This is done by choosing “Add > Selection Box” . Click on “Automatic” and a selection box will surround the symbol with pins on its boundary. The inverter symbol will look something like this.



Try to edit the symbol shape such that all its pin-names are within the selection box to avoid confusion later on when using the symbol for a higher level of design.

Basic idea about layout

- 1. Think of layout as 3-dimentional description for related circuits. You put all needed elements on silicon substrate, then connected with each other through metal layer, which means, you have to use "contact" to build connection with metal first, and then establish connection on metal layer.***
- 2. Metal layers are used for connection; make sure there is no crossing within same layer.***
- 3. There are limited contacts, including m1-p, m1-n, m1-poly, m2-m1, m3- m2, m4-m3 and m5-m4.***
- 4. Remember to put all your pins on metal layer.***
- 5. The background of your layout refers to p-substrate***
- 6. Make sure that your transistor "substrate" is connected to the "source".***
- 7. The width of PMOS should be at least double size of NMOS.***
- 8. Good procedure to draw layout is structure -> stick diagram -> layout***
- 9. Reference for how to efficiently draw a stick diagram:***
<http://www.ohio.edu/people/starzykj/webcad/ee415/VLSI/design/stick/stick.htm>